
Structure from Motion

Structure from motion

Given many images, how can we

- a) figure out where they were all taken from?
- b) build a 3D model of the scene?



This is the **structure from motion (SfM)** problem

Structure from motion

- **Input: images with 2D points in correspondence**
 $p_{i,j} = (u_{i,j}, v_{i,j})$
- **Output**
 - **structure:** 3D location x_i for each point p_i
 - **motion:** camera parameters R_j, t_j (and possibly intrinsic parameters too)
- **Objective function:** minimize *reprojection error*

Basic SfM Math Covered in Class

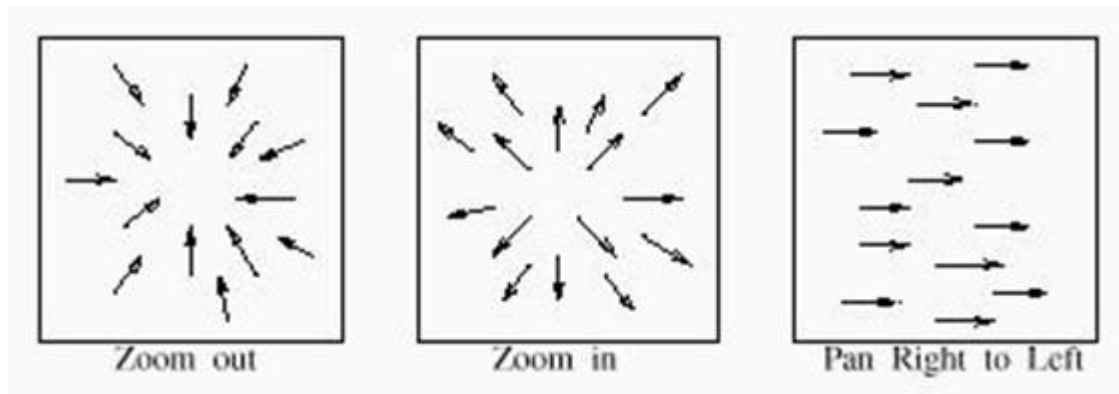
SfM Equations

$$u = \frac{T_Z x - T_X f}{Z} + \omega_X \left(\frac{xy}{f} \right) - \omega_Y \left(f + \frac{x^2}{f} \right) + \omega_Z y$$

$$v = \frac{T_Z y - T_Y f}{Z} + \omega_X \left(f + \frac{y^2}{f} \right) - \omega_Y \left(\frac{xy}{f} \right) + \omega_Z x$$

Notation defined in class

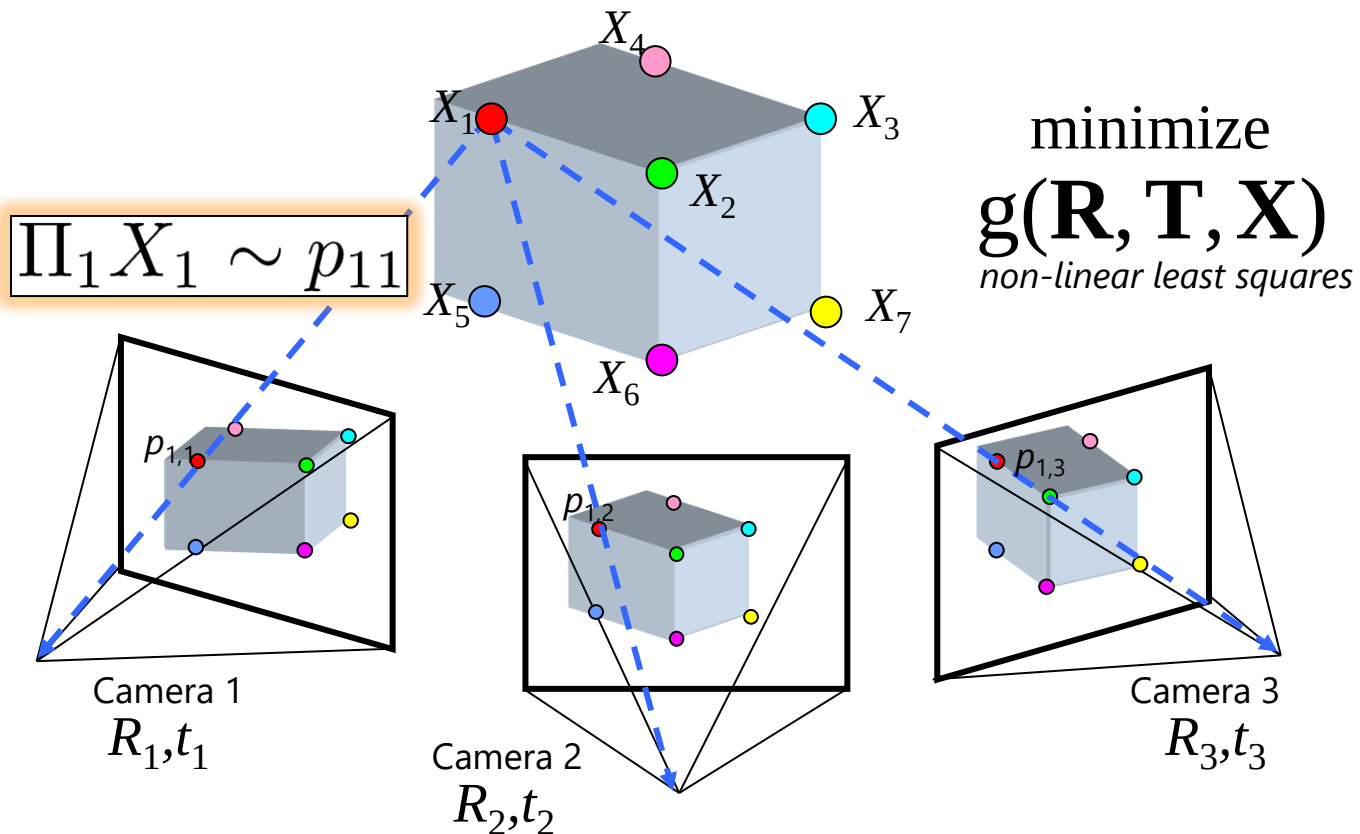
Focus of Expansion/Contraction



FOE/FOC

$T_z = 0$, FOE at infinity

Reprojection Error



Bundle Adjustment

- Minimize sum of squared reprojection errors:

$$g(\mathbf{X}, \mathbf{R}, \mathbf{T}) = \sum_{i=1}^m \sum_{j=1}^n \underbrace{w_{ij}}_{\substack{\text{indicator variable:} \\ \text{is point } i \text{ visible in image } j?}} \cdot \left\| \underbrace{\mathbf{P}(\mathbf{x}_i, \mathbf{R}_j, \mathbf{t}_j)}_{\substack{\text{predicted} \\ \text{image location}}} - \underbrace{\begin{bmatrix} u_{i,j} \\ v_{i,j} \end{bmatrix}}_{\substack{\text{observed} \\ \text{image location}}} \right\|^2$$

- Minimizing this function is called *bundle adjustment*
 - Optimized using non-linear least squares, e.g. Levenberg-Marquardt algorithm

Structure from motion: limitations

- If nothing is known (only images are available), there is an arbitrary projective ambiguity.
- Even with fully calibrated camera, we cannot recover the absolute scale.
- Need good feature tracker